

Project Title: AI Based Email Phishing Detection System

ABSTRACT

Phishing attacks are one of the most common cybersecurity threats, where attackers use fake emails to impersonate trusted organizations and steal sensitive information such as usernames, passwords, banking details, and personal data. As phishing techniques become more advanced, traditional rule-based detection methods often fail to identify new attacks. The **AI-Based Phishing Email Detection System** addresses this challenge by using Artificial Intelligence and Machine Learning to accurately classify emails as **Phishing** or **Legitimate**.

The system analyzes various email features, including the sender's address, email content, embedded URLs, and suspicious keywords. These features are processed using a trained machine learning model to predict whether an email is safe or malicious. It also provides a confidence score, highlights suspicious elements, and offers security recommendations to help users understand potential threats.

The application is built using **HTML, CSS, JavaScript, and Bootstrap** for the frontend, while **Python and Flask** are used for the backend. Libraries such as **Scikit-learn, Pandas, and NumPy** are used for data preprocessing, feature extraction, model training, and prediction. By combining AI with cybersecurity, the system improves email security, reduces the risk of phishing attacks, and promotes safer email usage through intelligent threat detection and user awareness.

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